Student Name: Hafsa Tariq

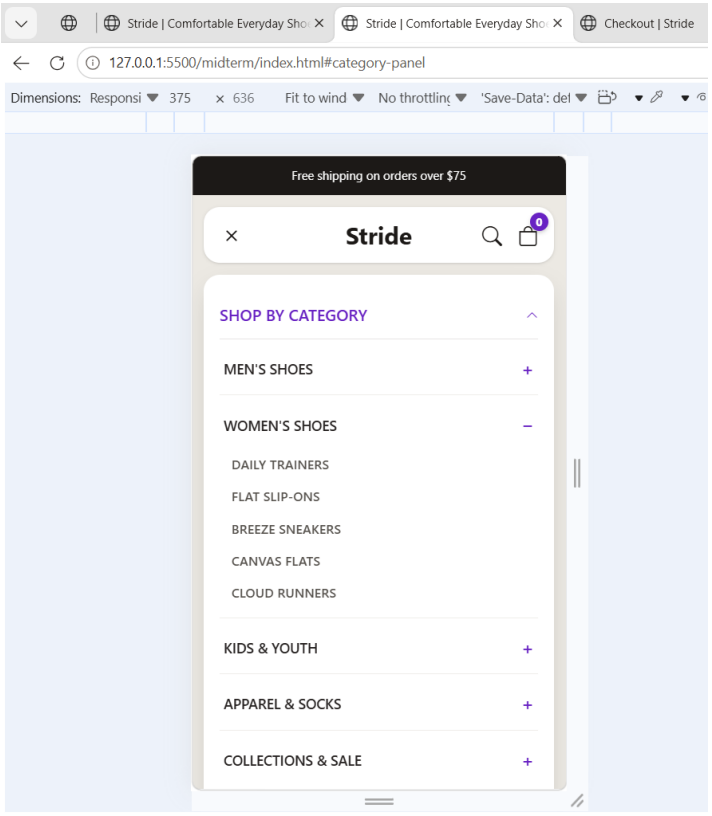
Registration Number: FA24-BCS-038

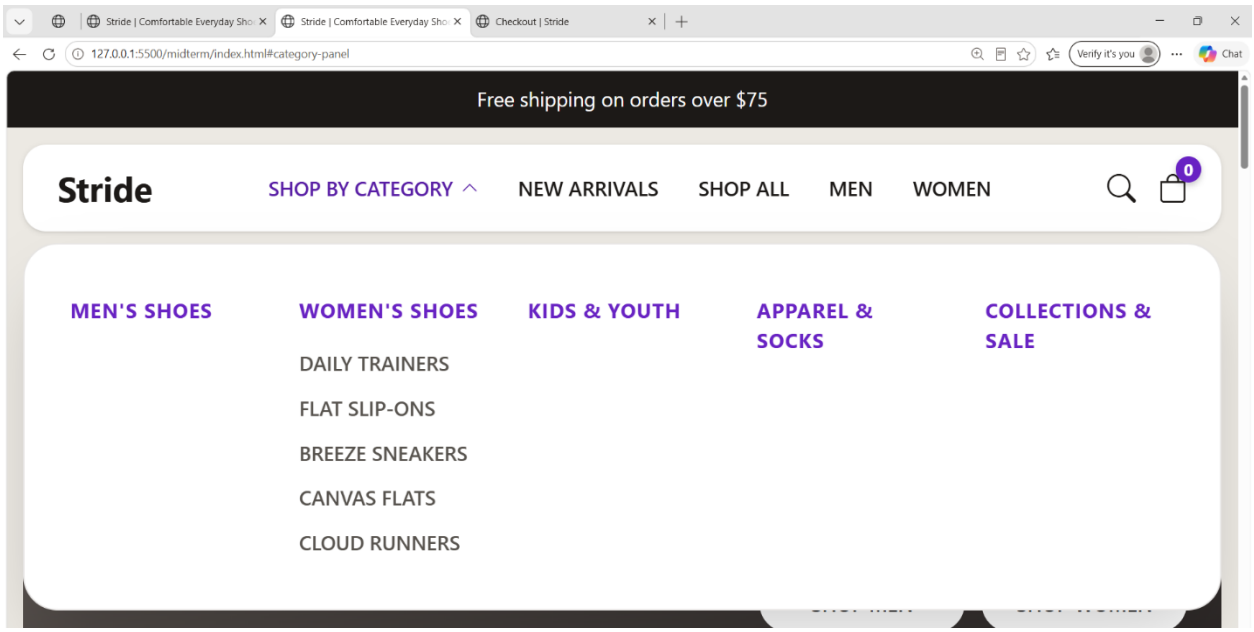
Subject: Web Technologies

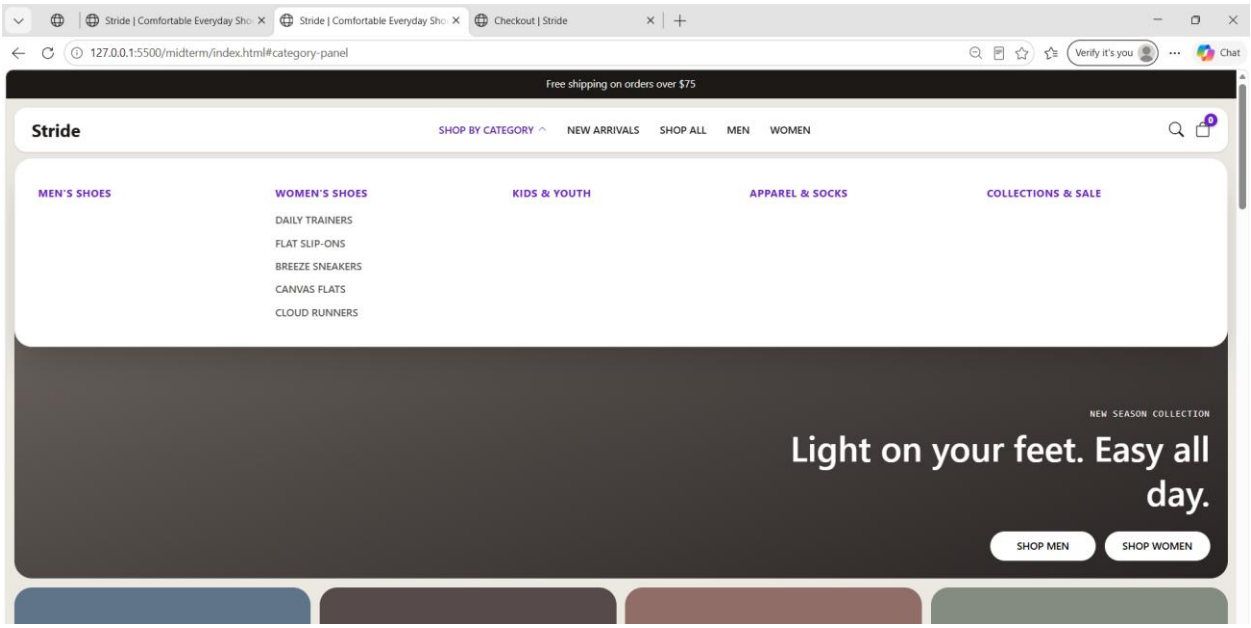
Personal Derived Numbers:

- Roll Number: FA24-BCS-038
- N (Last two digits): 38
- Breakpoint (BP = 600 + N): 638px
- Accent Hue (H = (38 × 7) mod 360): 266 -> Accent Colour: hsl(266, 70%, 45%)
- K ((38 mod 3) + 3): 5 columns / link groups
- Item Limit ((38 mod 4) + 2): 4 items per product
- Delivery Date Range: 2026-10-29 to 2026-12-06

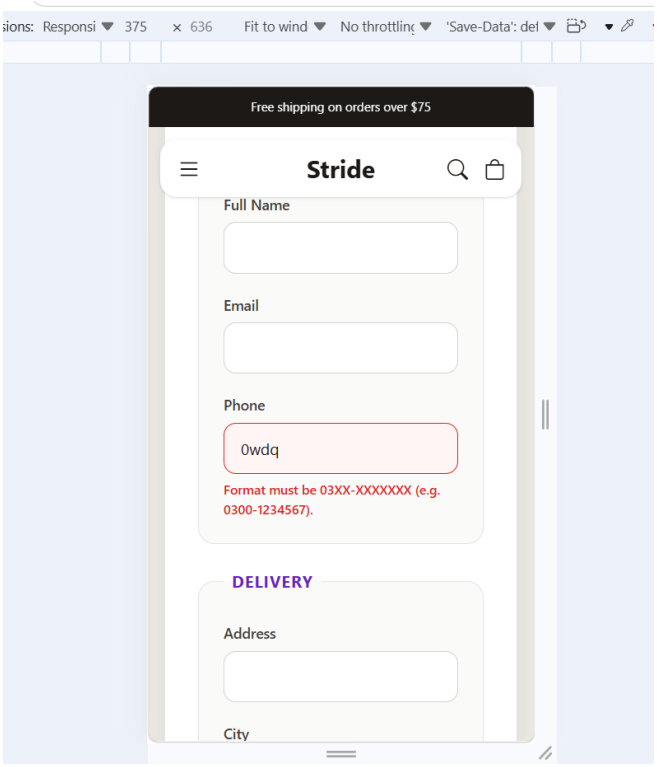
Task 1:

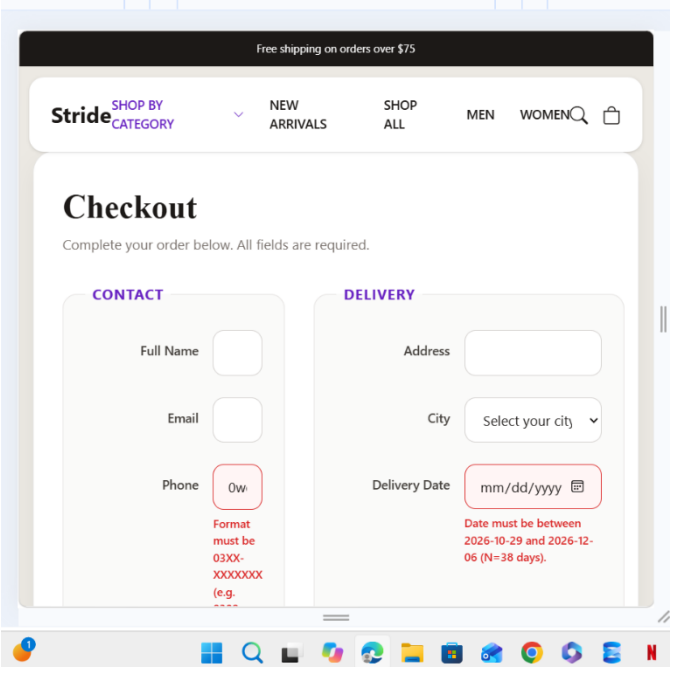


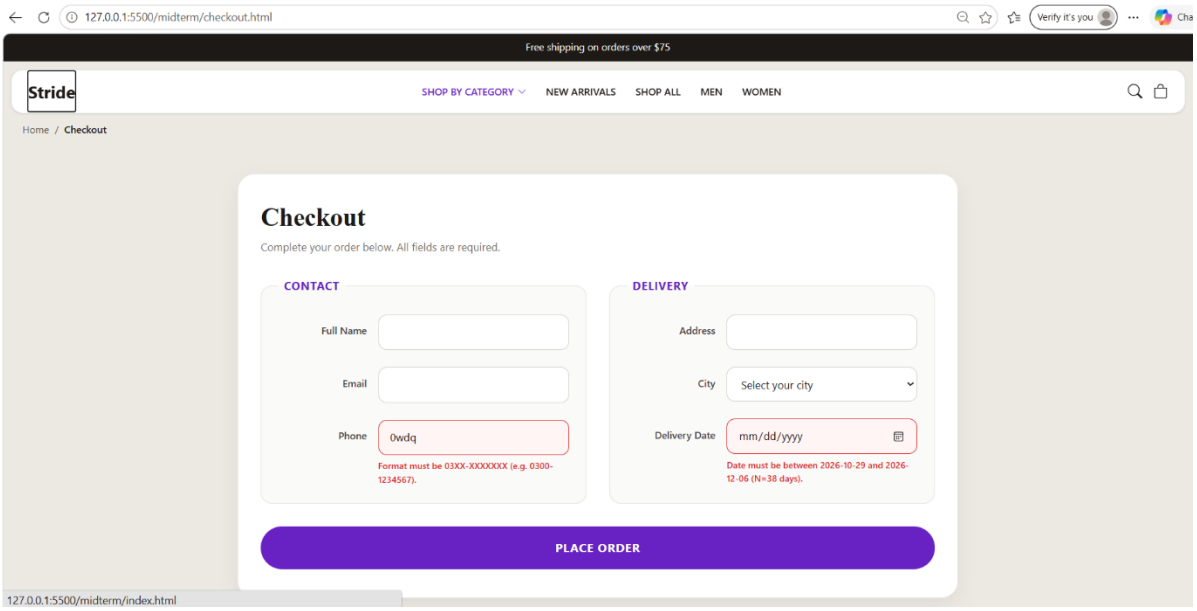

Stride | Comfortable Everyday Sho
Stride | Comfortable Everyday Sho
Checkout | Stride
127.0.0.1:5500/midterm/index.html#category-panel
Verify it's you
Chat
Free shipping on orders over $75
Stride
SHOP BY CATEGORY
NEW ARRIVALS
SHOP ALL
MEN
WOMEN
0
MEN'S SHOES
WOMEN'S SHOES
DAILY TRAINERS
FLAT SLIP-ONS
BREEZE SNEAKERS
CANVAS FLATS
CLOUD RUNNERS
KIDS & YOUTH
APPAREL & SOCKS
COLLECTIONS & SALE


Stride | Comfortable Everyday Sho
Stride | Comfortable Everyday Sho
Checkout | Stride
127.0.0.1:5500/midterm/index.html#category-panel
Verify it's you
Chat
Free shipping on orders over $75
Stride
SHOP BY CATEGORY
NEW ARRIVALS
SHOP ALL
MEN
WOMEN
0
MEN'S SHOES
WOMEN'S SHOES
DAILY TRAINERS
FLAT SLIP-ONS
BREEZE SNEAKERS
CANVAS FLATS
CLOUD RUNNERS
KIDS & YOUTH
APPAREL & SOCKS
COLLECTIONS & SALE
NEW SEASON COLLECTION
Light on your feet. Easy all day.
SHOP MEN
SHOP WOMEN

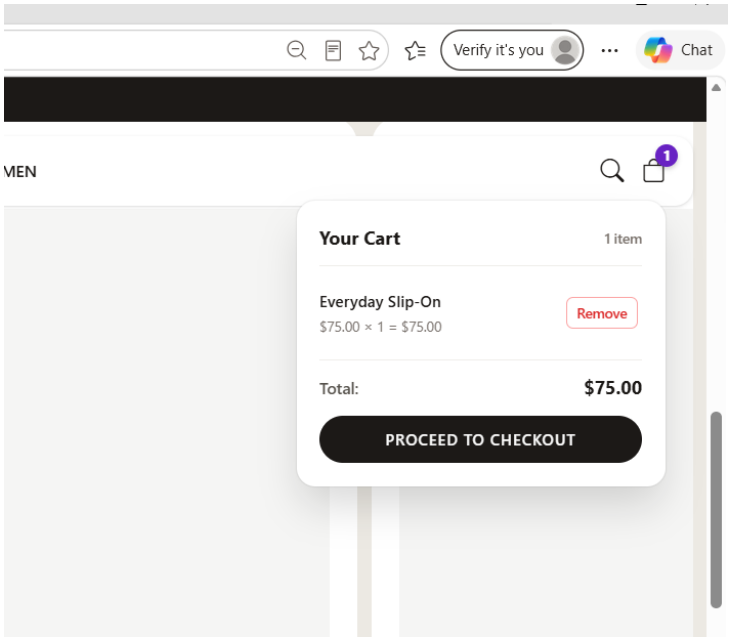
Task2:

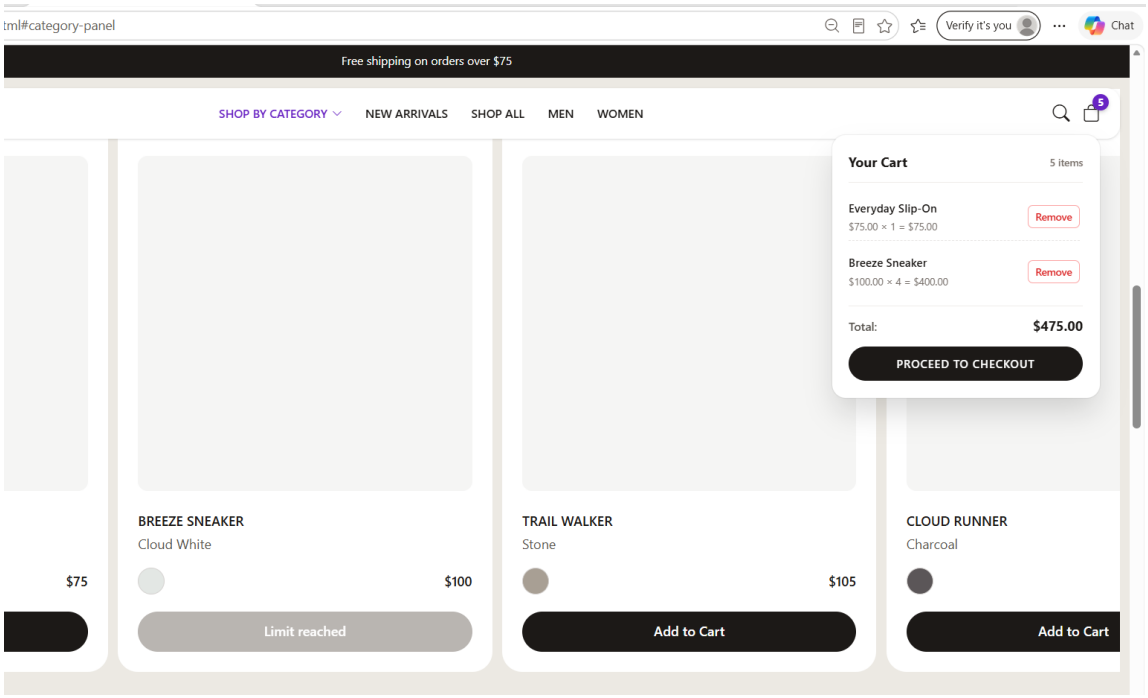

Free shipping on orders over $75
Stride
Full Name
Email
Phone
0wdq
Format must be 03XX-XXXXXXX (e.g. 0300-1234567).
DELIVERY
Address
City

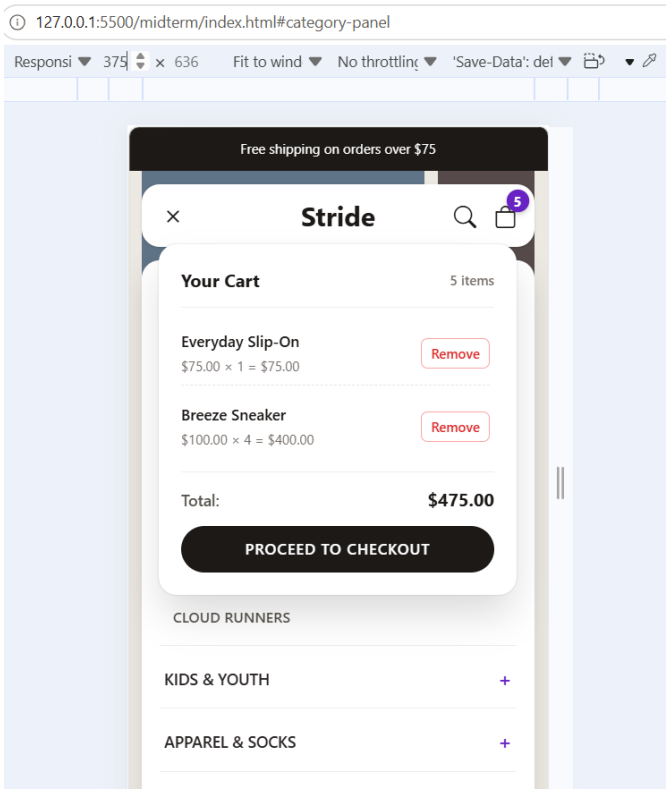

Free shipping on orders over $75
Stride
SHOP BY CATEGORY
NEW ARRIVALS
SHOP ALL
MEN
WOMEN
Checkout
Complete your order below. All fields are required.
CONTACT
Full Name
Email
Phone
0w
Format must be 03XX-XXXXXXX (e.g.
DELIVERY
Address
City
Select your city
Delivery Date
mm/dd/yyyy
Date must be between 2026-10-29 and 2026-12-06 (N=38 days).


127.0.0.1:5500/midterm/checkout.html
Verify it's you
Free shipping on orders over $75
Stride
SHOP BY CATEGORY
NEW ARRIVALS
SHOP ALL
MEN
WOMEN
Home / Checkout
Checkout
Complete your order below. All fields are required.
CONTACT
Full Name
Email
Phone
0wdq
Format must be 03XX-XXXXXXX (e.g. 0300-1234567).
DELIVERY
Address
City
Select your city
Delivery Date
mm/dd/yyyy
Date must be between 2026-10-29 and 2026-12-06 (N=38 days).
PLACE ORDER
127.0.0.1:5500/midterm/index.html

Task 3:


Verify it's you
Chat
MEN
1
Your Cart
1 item
Everyday Slip-On
$75.00 × 1 = $75.00
Remove
Total:
$75.00
PROCEED TO CHECKOUT


tml#category-panel
Verify it's you
Chat
Free shipping on orders over $75
SHOP BY CATEGORY
NEW ARRIVALS
SHOP ALL
MEN
WOMEN
5
Your Cart
5 items
Everyday Slip-On
$75.00 × 1 = $75.00
Remove
Breeze Sneaker
$100.00 × 4 = $400.00
Remove
Total:
$475.00
PROCEED TO CHECKOUT
$75
BREEZE SNEAKER
Cloud White
$100
Limit reached
TRAIL WALKER
Stone
$105
Add to Cart
CLOUD RUNNER
Charcoal
Add to Cart


127.0.0.1:5500/midterm/index.html#category-panel
Responsi
375
× 636
Fit to wind
No throttlin
'Save-Data': del
Free shipping on orders over $75
Stride
5
Your Cart
5 items
Everyday Slip-On
$75.00 × 1 = $75.00
Remove
Breeze Sneaker
$100.00 × 4 = $400.00
Remove
Total:
$475.00
PROCEED TO CHECKOUT
CLOUD RUNNERS
KIDS & YOUTH
APPAREL & SOCKS

Task 1:

```
<!-- Task 1: Shop by Category Mega-Menu (K = 5, BP = 638px) -->
<div class="category-menu-item">
 <a href="#category-panel" class="category-menu-btn nav-bp-link" tabindex="0">
  Shop by Category
  <i class="bi bi-chevron-down category-btn-icon"></i>
 </a>

 <div id="category-panel" class="category-mega-panel">
  <div class="category-columns-grid">

   <!-- Column 1 -->
   <details class="category-col">
    <summary class="category-col-heading">Men's Shoes</summary>
    <ul class="category-links-list">
     <li><a href="#">Running Shoes</a></li>
     <li><a href="#">Everyday Slip-Ons</a></li>
     <li><a href="#">Leather Loafers</a></li>
     <li><a href="#">Trail Walkers</a></li>
     <li><a href="#">All-Weather Boots</a></li>
    </ul>
   </details>

   <!-- Column 2 -->
   <details class="category-col">
    <summary class="category-col-heading">Women's Shoes</summary>
```

```
 <ul class="category-links-list">
  <li><a href="#">Daily Trainers</a></li>
  <li><a href="#">Flat Slip-Ons</a></li>
  <li><a href="#">Breeze Sneakers</a></li>
  <li><a href="#">Canvas Flats</a></li>
  <li><a href="#">Cloud Runners</a></li>
 </ul>
</details>

<!-- Column 3 -->
<details class="category-col">
 <summary class="category-col-heading">Kids & Youth</summary>
 <ul class="category-links-list">
  <li><a href="#">School Everyday</a></li>
  <li><a href="#">Sport Runners</a></li>
  <li><a href="#">Slip-On Play</a></li>
  <li><a href="#">Waterproof Boots</a></li>
  <li><a href="#">Toddler First Steps</a></li>
 </ul>
</details>

<!-- Column 4 -->
<details class="category-col">
 <summary class="category-col-heading">Apparel & Socks</summary>
 <ul class="category-links-list">
  <li><a href="#">Ankle Socks</a></li>
```

```
        <li><a href="#">Crew Running Socks</a></li>
        <li><a href="#">Merino Wool Tees</a></li>
        <li><a href="#">Relaxed Sweatpants</a></li>
        <li><a href="#">Organic Caps</a></li>
      </ul>
    </details>

    <!-- Column 5 -->
    <details class="category-col">
      <summary class="category-col-heading">Collections & Sale</summary>
      <ul class="category-links-list">
        <li><a href="#">New Season Arrivals</a></li>
        <li><a href="#">Bestsellers Pack</a></li>
        <li><a href="#">Limited Editions</a></li>
        <li><a href="#">Sustainable Series</a></li>
        <li><a href="#">Clearance Sale</a></li>
      </ul>
    </details>

   </div>
  </div>
</div>
```

CSS Styles:

```
/* Responsive Mega-Menu Styles (Breakpoint BP = 638px) */
:root {
  --midterm-accent: hsl(266, 70%, 45%);
```

```
}

/* Mobile Accordion View (< 638px) */
@media (max-width: 637.98px) {
  .category-mega-panel {
    display: block;
    width: 100%;
  }
  .category-columns-grid {
    display: flex;
    flex-direction: column;
    gap: 0.5rem;
    padding-left: 0.5rem;
  }
  .category-col-heading {
    cursor: pointer;
    font-weight: 600;
    font-size: 0.875rem;
    padding: 0.5rem 0;
    list-style: none;
    display: flex;
    justify-content: space-between;
  }
}

/* Desktop Floating Mega-Panel (>= 638px) */
```

```
@media (min-width: 638px) {
  .category-mega-panel {
    display: none;
    position: absolute;
    left: 0.75rem;
    right: 0.75rem;
    top: calc(100% - 0.25rem);
    z-index: 100;
  }
  .category-menu-item:hover .category-mega-panel,
  .category-menu-item:focus-within .category-mega-panel {
    display: block;
  }
  .category-columns-grid {
    display: grid;
    grid-template-columns: repeat(5, minmax(0, 1fr));
    gap: 1.75rem;
    background-color: #ffffff;
    border-radius: 1.5rem;
    padding: 2.25rem 2rem;
    box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.2);
  }
  .category-col:not([open]) > :not(summary) {
    display: block !important;
  }
}
```

Task2:

```
<form id="checkout-form" class="checkout-form-grid" action="#" method="POST" novalidate="false">


 <!-- Fieldset 1: Contact -->

 <fieldset class="checkout-fieldset">

  <legend class="checkout-legend">Contact</legend>


  <div class="form-group">

   <label for="fullname">Full Name</label>

   <input type="text" id="fullname" name="fullname" required placeholder="John Doe" autocomplete="name">

   <span class="field-error-msg">Please enter your full name.</span>

  </div>


  <div class="form-group">

   <label for="email">Email</label>

   <input type="email" id="email" name="email" required placeholder="name@example.com" autocomplete="email">

   <span class="field-error-msg">Please enter a valid email address.</span>

  </div>


  <div class="form-group">

   <label for="phone">Phone</label>

   <input type="tel" id="phone" name="phone" required pattern="03[0-9]{2}-[0-9]{7}" placeholder="0300-1234567" autocomplete="tel">
```

```
    <span class="field-error-msg">Format must be 03XX-XXXXXXX (e.g. 0300-1234567).</span>
   </div>
  </fieldset>

  <!-- Fieldset 2: Delivery -->
  <fieldset class="checkout-fieldset">
   <legend class="checkout-legend">Delivery</legend>

   <div class="form-group">
    <label for="address">Address</label>
    <input type="text" id="address" name="address" required placeholder="123 Street Name, Area" autocomplete="street-address">
    <span class="field-error-msg">Please enter your delivery address.</span>
   </div>

   <div class="form-group">
    <label for="city">City</label>
    <select id="city" name="city" required>
     <option value="" disabled selected hidden>Select your city</option>
     <option value="Islamabad">Islamabad</option>
     <option value="Rawalpindi">Rawalpindi</option>
     <option value="Lahore">Lahore</option>
     <option value="Karachi">Karachi</option>
     <option value="Peshawar">Peshawar</option>
    </select>
    <span class="field-error-msg">Please select a city.</span>
```

```
  </div>

  <div class="form-group">
   <label for="delivery-date">Delivery Date</label>
   <input type="date" id="delivery-date" name="delivery_date" required min="2026-10-29" max="2026-12-06">
   <span class="field-error-msg">Date must be between 2026-10-29 and 2026-12-06 (N=38 days).</span>
  </div>
 </fieldset>

 <!-- Submit button: Full width -->
 <div class="checkout-submit-wrapper">
  <button type="submit" class="checkout-submit-btn">Place Order</button>
 </div>

</form>
```

CSS:

```
/* Interactive HTML5 Validation Styling */
.checkout-fieldset input:user-invalid,
.checkout-fieldset select:user-invalid,
.checkout-fieldset input:not(:placeholder-shown):invalid {
 border-color: #ef4444;
 background-color: #fef2f2;
}

.checkout-fieldset input:user-invalid ~ .field-error-msg,
```

```
.checkout-fieldset select:user-invalid ~ .field-error-msg,
.checkout-fieldset input:not(:placeholder-shown):invalid ~ .field-error-msg {
  display: block;
  color: #dc2626;
  font-size: 0.75rem;
  margin-top: 0.25rem;
}

/* Responsive Grid Layout (BP = 638px) */
@media (min-width: 638px) {
  .checkout-form-grid {
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 1.75rem;
  }
  .form-group {
    display: grid;
    grid-template-columns: 140px 1fr;
    align-items: center;
    column-gap: 1rem;
  }
}

@media (max-width: 637.98px) {
  .checkout-form-grid {
    grid-template-columns: 1fr;
  }
```

```
  .form-group {
    display: flex;
    flex-direction: column;
    gap: 0.35rem;
  }
}
```

Task3:

```
// Midterm Task 3: Working cart badge and mini cart
// Roll Number: FA24-BCS-038 (N = 38)
// Limit formula: (N mod 4) + 2 = (38 mod 4) + 2 = 2 + 2 = 4
const MAX_ITEM_LIMIT = (38 % 4) + 2; // Exactly 4 items per product
const cart = [];

function initCart() {
  const cartBtn = document.getElementById('cart-btn');
  const cartBadge = document.getElementById('cart-badge');
  const miniCartPanel = document.getElementById('mini-cart-panel');
  const miniCartItems = document.getElementById('mini-cart-items');
  const miniCartCount = document.getElementById('mini-cart-count');
  const miniCartTotal = document.getElementById('mini-cart-total');

  // Find all product cards on the listing page
  const productCards = document.querySelectorAll('article');

  // Attach event listeners to "Add to Cart" buttons
  productCards.forEach((card) => {
```

```
const btn = card.querySelector('button');
const nameEl = card.querySelector('h3');

if (!btn || !nameEl) return;

btn.addEventListener('click', (e) => {
 e.preventDefault();

 // Read product name from the card DOM
 const name = nameEl.textContent.trim();

 // Read price from the card DOM (looks for $ sign)
 let price = 0;
 const allTextElements = card.querySelectorAll('span, p, div');
 for (const el of allTextElements) {
  const text = el.textContent.trim();
  if (text.startsWith('$')) {
   price = parseFloat(text.replace(/[^0-9.]/g, '')) || 0;
   break;
  }
 }

 // Check current quantity in cart
 const currentCount = cart.filter((item) => item.name === name).length;

 if (currentCount < MAX_ITEM_LIMIT) {
```

```
      cart.push({
        id: Date.now() + Math.random(),
        name: name,
        price: price,
      });
      updateCartUI();
    }
  });
});

// Calculate total price using Array.prototype.reduce()
function calculateTotal() {
  return cart.reduce((total, item) => total + item.price, 0);
}

// Update badge, mini-cart list, total, and card buttons
function updateCartUI() {
  const totalCount = cart.length;

  // 1. Update Cart Badge
  if (cartBadge) {
    cartBadge.textContent = totalCount;
  }

  // 2. Update Mini-Cart Count
  if (miniCartCount) {
```

```
  miniCartCount.textContent = `${totalCount} item${totalCount === 1 ? '' : 's'}`;
}

// 3. Update Total using reduce()
if (miniCartTotal) {
  const total = calculateTotal();
  miniCartTotal.textContent = `$${total.toFixed(2)}`;
}

// 4. Render Mini-Cart Items list
if (miniCartItems) {
  if (cart.length === 0) {
    miniCartItems.innerHTML = '<p class="empty-cart-msg">Your cart is empty.</p>';
  } else {
    const grouped = {};
    cart.forEach((item) => {
      if (!grouped[item.name]) {
        grouped[item.name] = { name: item.name, price: item.price, count: 0 };
      }
      grouped[item.name].count += 1;
    });

    miniCartItems.innerHTML = '';
    Object.values(grouped).forEach((grp) => {
      const itemRow = document.createElement('div');
      itemRow.className = 'mini-cart-item';
```

```
      itemRow.innerHTML = `
       <div class="mini-cart-item-info">
        <p class="mini-cart-item-name">${grp.name}</p>
        <p class="mini-cart-item-meta">$${grp.price.toFixed(2)} × ${grp.count} = $${(grp.price * grp.count).toFixed(2)}</p>
       </div>
       <button type="button" class="mini-cart-remove-btn" aria-label="Remove one ${grp.name}">Remove</button>
      `;

      // Remove button handler
      const removeBtn = itemRow.querySelector('.mini-cart-remove-btn');
      removeBtn.addEventListener('click', () => {
       removeOneProduct(grp.name);
      });

      miniCartItems.appendChild(itemRow);
     });
    }
  }

  // 5. Update Product Card Buttons (Limit reached state)
  productCards.forEach((card) => {
   const btn = card.querySelector('button');
   const nameEl = card.querySelector('h3');
   if (!btn || !nameEl) return;
```

```
    const name = nameEl.textContent.trim();
    const countInCart = cart.filter((item) => item.name === name).length;

    if (countInCart >= MAX_ITEM_LIMIT) {
      btn.textContent = 'Limit reached';
      btn.disabled = true;
      btn.classList.add('limit-reached');
    } else {
      btn.textContent = 'Add to Cart';
      btn.disabled = false;
      btn.classList.remove('limit-reached');
    }
  });
}

// Remove ONE item from cart
function removeOneProduct(productName) {
  let indexToRemove = -1;
  for (let i = cart.length - 1; i >= 0; i--) {
    if (cart[i].name === productName) {
      indexToRemove = i;
      break;
    }
  }
  if (indexToRemove !== -1) {
    cart.splice(indexToRemove, 1);
```

```
    updateCartUI();
  }
 }

 // Toggle mini-cart panel on cart icon click
 if (cartBtn && miniCartPanel) {
  cartBtn.addEventListener('click', (e) => {
   e.preventDefault();
   miniCartPanel.classList.toggle('hidden');
  });

  document.addEventListener('click', (e) => {
   if (!cartBtn.contains(e.target) && !miniCartPanel.contains(e.target)) {
    miniCartPanel.classList.add('hidden');
   }
  });
 }

 // Initial sync
 updateCartUI();
}

document.addEventListener('DOMContentLoaded', initCart);
```